

TrES-3b

R-2757

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130628 · created 2026-09-03T00:07:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456471.736 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.057 +/- 0.054
Transit depth (Rp/Rs)^2	0.33 +/- 0.62 [%]
Orbital Inclination (inc)	82.21 +/- 2.91
Airmass coefficient 1 (a1)	1.81 +/- 0.12
Airmass coefficient 2 (a2)	-0.57 +/- 0.12
Scatter in the residuals of the lightcurve fit is	6.8 %
Best Comparison Star	#3 - [555, 76]
Optimal Method	PSF photometry
Transit Duration (day)	0.04 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.057 ± 0.054 vs 0.16309 (deviation 1.48σ)
Duration fit vs published	0.04 d vs 0.0567 d (deviation 0.46σ)
Mid-time O-C	-2.3 min
Depth SNR	0.8
Residual scatter	6.8 %

Light curve

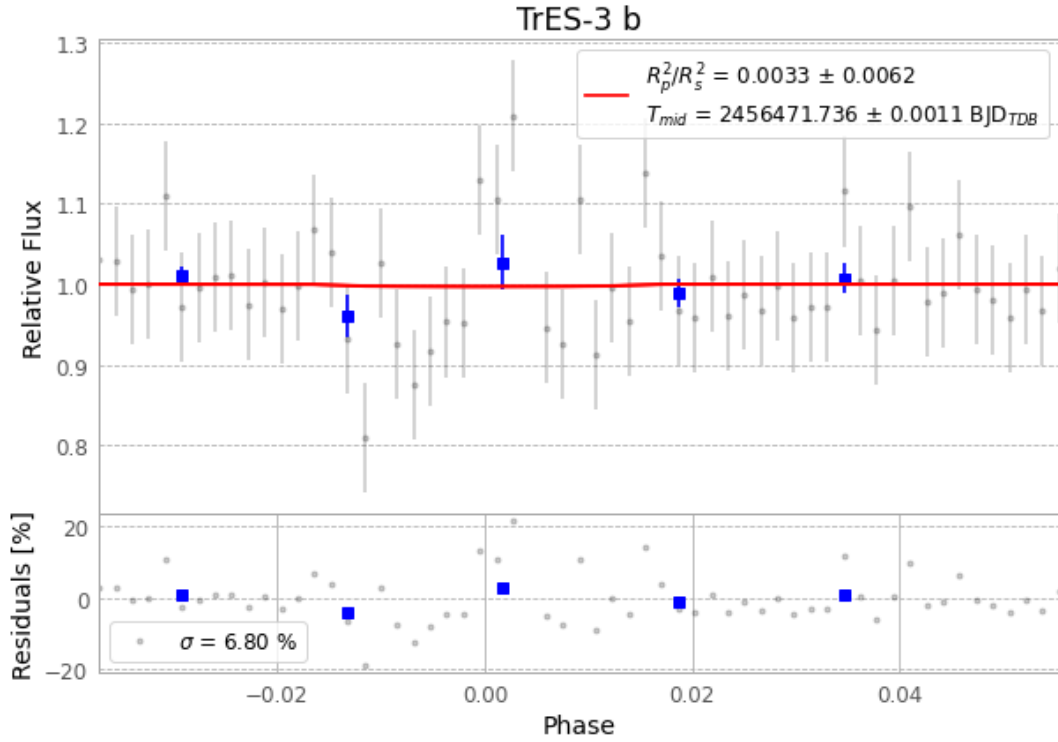


Figure 1 — FinalLightCurve_TrES-3 b_2013-06-27.png (SHA-256 a58a084c1e8dff6d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7d2b610f20525ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), TRES-3130628042412.FITS → TRES-3130628071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130628.log (9c8ad8c4fdaf...), FinalLightCurve_TrES-3 b_2013-06-27.png (a58a084c1e8d...), FinalLightCurve_TrES-3 b_2013-06-27.csv (b3eb3b3050fc...)

Compute resources

Wall time 116.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:07Z.